MA1522 Linear Algebra for Computing
L2 Lesson 3



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Semester 1, 2026/27

Matrix Operations

1. **Matrix Addition.**
   Let $\boldsymbol{A} = (a_{ij})_{m\times n}$ and $\boldsymbol{B} = (b_{ij})_{m\times n}$. Then
   $$\boldsymbol{A} + \boldsymbol{B} = (a_{ij} + b_{ij})_{m\times n}.$$
2. **Scalar Multiplication.**
   Let $A = (a_{ij})_{m\times n}$ and $c \in \mathbb{R}$. Then
   $$c\boldsymbol{A} = (c\, a_{ij})_{m\times n}.$$
3. **Matrix Multiplication.**
   Let $A = (a_{ij})_{m\times n}$ and $B = (b_{ij})_{n\times p}$. Then
   $$AB = \left(\sum_{k=1}^{n} a_{ik}b_{kj}\right)_{m\times p}.$$
4. **Matrix Transposition.**
   Let $A = (a_{ij})_{m\times n}$. Then
   $$A^T = (a_{ji})_{n\times m}.$$

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Properties of Matrix Operations

Let $\boldsymbol{A}$ be an $m \times n$ matrix and let $c \in \mathbb{R}$. Let $\boldsymbol{B}$ and $\boldsymbol{C}$ be matrices of sizes so that the operations below make sense.

1. $\boldsymbol{A} + \boldsymbol{B} = \boldsymbol{B} + \boldsymbol{A}$.
2. $(\boldsymbol{A} + \boldsymbol{B}) + \boldsymbol{C} = \boldsymbol{A} + (\boldsymbol{B} + \boldsymbol{C})$.
3. $(\boldsymbol{AB})\boldsymbol{C} = \boldsymbol{A}(\boldsymbol{BC})$.
4. $c(\boldsymbol{A} + \boldsymbol{B}) = c\boldsymbol{A} + c\boldsymbol{B}$.
5. $c(\boldsymbol{AB}) = (c\boldsymbol{A})\boldsymbol{B} = \boldsymbol{A}(c\boldsymbol{B})$.
6. $\boldsymbol{A}(\boldsymbol{B} + \boldsymbol{C}) = \boldsymbol{AB} + \boldsymbol{AC}$.
7. $(\boldsymbol{A} + \boldsymbol{B})\boldsymbol{C} = \boldsymbol{AC} + \boldsymbol{BC}$.
8. $\boldsymbol{A}\boldsymbol{I}_n = \boldsymbol{A} = \boldsymbol{I}_m\boldsymbol{A}$.
9. $\boldsymbol{A} + \boldsymbol{0}_{m\times n} = \boldsymbol{A} = \boldsymbol{0}_{m\times n} + \boldsymbol{A}$.
10. $\boldsymbol{A}\boldsymbol{0}_{n\times n} = \boldsymbol{0}_{m\times n} = \boldsymbol{0}_{m\times m}\boldsymbol{A}$.

Properties of Matrix Operations (Cont'd)

11. $(\boldsymbol{A}+\boldsymbol{B})^T = \boldsymbol{A}^T + \boldsymbol{B}^T$.
12. $(c\boldsymbol{A})^T = c(\boldsymbol{A}^T)$.
13. $(\boldsymbol{A}\boldsymbol{B})^T = \boldsymbol{B}^T\boldsymbol{A}^T$.

Non-properties of Matrix Operations

1. $\boldsymbol{AB} \neq \boldsymbol{BA}$.
2. $(\boldsymbol{AB})^T \neq \boldsymbol{A}^T\boldsymbol{B}^T$.
3. $\boldsymbol{AB} = \boldsymbol{0} \not\Rightarrow \boldsymbol{A} = \boldsymbol{0}$ or $\boldsymbol{B} = \boldsymbol{0}$.

Question 1

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Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be square matrices of order n. Which of the following statements is/are (always) true?

a. $(\boldsymbol{A}+\boldsymbol{I}_n)^2 = \boldsymbol{A}^2 + 2\boldsymbol{A} + \boldsymbol{I}_n$.

b. $(\boldsymbol{A}+\boldsymbol{B})^2 = \boldsymbol{A}^2 + 2\boldsymbol{A}\boldsymbol{B} + \boldsymbol{B}^2$.

c. If $\boldsymbol{A}^2 = \boldsymbol{B}^2$, then $\boldsymbol{A} = \boldsymbol{B}$ or $\boldsymbol{A} = -\boldsymbol{B}$.

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c. If $\boldsymbol{A}^2 = \boldsymbol{B}^2$, then $\boldsymbol{A} = \boldsymbol{B}$ or $\boldsymbol{A} = -\boldsymbol{B}$. ..... **Ans: False**

Question 1(a): Solution

a. $(\boldsymbol{A}+\boldsymbol{I}_n)^2=\boldsymbol{A}^2+2\boldsymbol{A}+\boldsymbol{I}_n$.

$$
\begin{aligned}
(\boldsymbol{A}+\boldsymbol{I}_n)^2 &= (\boldsymbol{A}+\boldsymbol{I}_n)(\boldsymbol{A}+\boldsymbol{I}_n)\\
&= (\boldsymbol{A}+\boldsymbol{I}_n)\boldsymbol{A}+(\boldsymbol{A}+\boldsymbol{I}_n)\boldsymbol{I}_n\\
&= \boldsymbol{A}^2+\boldsymbol{I}_n\boldsymbol{A}+\boldsymbol{A}\boldsymbol{I}_n+\boldsymbol{I}_n^2\\
&= \boldsymbol{A}^2+\boldsymbol{A}+\boldsymbol{A}+\boldsymbol{I}_n^2\\
&= \boldsymbol{A}^2+2\boldsymbol{A}+\boldsymbol{I}_n.
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Question 1(a): Solution

a. $(\boldsymbol{A}+\boldsymbol{I}_n)^2 = \boldsymbol{A}^2 + 2\boldsymbol{A} + \boldsymbol{I}_n$.

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(\boldsymbol{A}+\boldsymbol{I}_n)^2 &= (\boldsymbol{A}+\boldsymbol{I}_n)(\boldsymbol{A}+\boldsymbol{I}_n) \\
&= (\boldsymbol{A}+\boldsymbol{I}_n)\boldsymbol{A} + (\boldsymbol{A}+\boldsymbol{I}_n)\boldsymbol{I}_n \\
&= \boldsymbol{A}^2 + \boldsymbol{I}_n\boldsymbol{A} + \boldsymbol{A}\boldsymbol{I}_n + \boldsymbol{I}_n^2 \\
&= \boldsymbol{A}^2 + \boldsymbol{A} + \boldsymbol{A} + \boldsymbol{I}_n^2 \\
&= \boldsymbol{A}^2 + 2\boldsymbol{A} + \boldsymbol{I}_n.
\end{aligned}
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Question 1(b): Solution

b. $(A+B)^2 = A^2 + 2AB + B^2$.

Similar to part (a),

$$\begin{aligned}(A+B)^2 &= A^2 + BA + AB + B^2 \\ &\neq A^2 + AB + AB + B^2 = A^2 + 2AB + B^2.\end{aligned}$$

Specific counter-example: $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Then

$$(A+B)^2 = \left(\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\right)^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

$$\begin{aligned}A^2 + 2AB + B^2 &= \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}^2 + 2\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}^2 \\ &= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} + 2\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}.\end{aligned}$$

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b $(A+B)^2 = A^2 + 2AB + B^2$.

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Similar to part (a),

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Specific counter-example: $\boldsymbol{A} = \left(\begin{smallmatrix}0 & 1\\ 0 & 0\end{smallmatrix}\right)$, $\boldsymbol{B} = \left(\begin{smallmatrix}0 & 0\\ 1 & 0\end{smallmatrix}\right)$. Then

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$$\begin{aligned}\boldsymbol{A}^2 + 2\boldsymbol{AB} + \boldsymbol{B}^2 &= \left(\begin{smallmatrix}0 & 1\\ 0 & 0\end{smallmatrix}\right)^2 + 2\left(\begin{smallmatrix}0 & 1\\ 0 & 0\end{smallmatrix}\right)\left(\begin{smallmatrix}0 & 0\\ 1 & 0\end{smallmatrix}\right) + \left(\begin{smallmatrix}0 & 0\\ 1 & 0\end{smallmatrix}\right)^2 \\ &= \left(\begin{smallmatrix}0 & 0\\ 0 & 0\end{smallmatrix}\right) + 2\left(\begin{smallmatrix}1 & 0\\ 0 & 0\end{smallmatrix}\right) + \left(\begin{smallmatrix}0 & 0\\ 0 & 0\end{smallmatrix}\right) = \left(\begin{smallmatrix}2 & 0\\ 0 & 0\end{smallmatrix}\right).\end{aligned}$$

Question 1(c): Solution

- If $A^2 = B^2$, then $A = B$ or $A = -B$.

  If $a, b \in \mathbb{R}$ such that $a^2 = b^2$, then

  $$0 = a^2 - b^2 = (a+b)(a-b),$$

  so that $a + b = 0$ or $a - b = 0$, or equivalently, $a = -b$ or $a = b$.

  But $(A+B)(A-B) = A^2 + BA - AB - B^2 \neq A^2 - B^2$ since $AB \neq BA$.

  Furthermore $CD = \mathbf{0} \not\Rightarrow C = \mathbf{0}$ or $D = \mathbf{0}$.

  Specific counter-example: $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Then

  $$A^2 = \mathbf{0} = B^2.$$

  But $A \neq \pm B$.

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  But $(\boldsymbol{A} + \boldsymbol{B})(\boldsymbol{A} - \boldsymbol{B}) = \boldsymbol{A}^2 + \boldsymbol{B}\boldsymbol{A} - \boldsymbol{A}\boldsymbol{B} - \boldsymbol{B}^2 \neq \boldsymbol{A}^2 - \boldsymbol{B}^2$ since $\boldsymbol{A}\boldsymbol{B} \neq \boldsymbol{B}\boldsymbol{A}$.

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Matrix Equations

Consider the following linear system of m equations in n variables $x_1, x_2, \ldots, x_n$:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2, \\ \vdots \qquad\qquad \vdots \qquad\qquad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m. \end{cases}$$

This can be written in matrix form:

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b},$$

where $\boldsymbol{A} = (a_{ij})_{m\times n}$,

$$\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \ \boldsymbol{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}.$$

Question 2

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Let $\boldsymbol{A}$ be an $m \times n$ matrix and $\boldsymbol{B}$ be an $p \times m$ matrix.

a. If $m < n$, how many solution(s) does $\boldsymbol{Ax} = \boldsymbol{0}$ have?
   (i) Exactly one. (iii) Always infinitely many.
   (ii) At least one. (iv) None of the above.

b. If $m > n$, how many solution(s) does $\boldsymbol{Ax} = \boldsymbol{0}$ have?
   (i) Exactly one. (iii) Always infinitely many.
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c. If $\boldsymbol{Ax} = \boldsymbol{0}$ has infinitely many solutions, how many solution(s) does $\boldsymbol{BAx} = \boldsymbol{0}$ have?
   (i) Exactly one. (iii) Always infinitely many.
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d. If $\boldsymbol{BAx} = \boldsymbol{0}$ has exactly one solution, how many solution(s) does $\boldsymbol{Ax} = \boldsymbol{0}$ have?
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Question 2(a,b): Solution

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The augmented matrix of the system of equations represented by $\boldsymbol{Ax} = \mathbf{0}$ is $\begin{pmatrix} \boldsymbol{A} & | & \mathbf{0} \end{pmatrix}$. Let $\boldsymbol{R} = \mathrm{RREF}(\boldsymbol{A})$. Then $\begin{pmatrix} \boldsymbol{R} & | & \mathbf{0} \end{pmatrix} = \mathrm{RREF}\begin{pmatrix} \boldsymbol{A} & | & \mathbf{0} \end{pmatrix}$.

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In particular, the rightmost column of the RREF is non-pivot. Thus the system of linear equations is consistent, i.e. there exists solution(s). (Alternatively, $\boldsymbol{x} = \mathbf{0}$ is always a solution.)

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Whether $\boldsymbol{Ax} = \mathbf{0}$ has exactly one solution or infinitely many solutions depends on whether there are arbitrary parameters in the general solutuion, equivalently, whether there are non-pivot columns in $\boldsymbol{R}$.

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If $m > n$, then this depends on $\boldsymbol{R}$; eg $\boldsymbol{R} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$ or $\begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$.

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a. (iii) Always infinitely many.

b. (ii) At least one.

Question 2(c,d): Solution

Note that if $\boldsymbol{u}$ is a solution of $\boldsymbol{Ax} = \boldsymbol{0}$, then

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Note that if $\boldsymbol{u}$ is a solution of $\boldsymbol{Ax} = \boldsymbol{0}$, then

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Thus every solution of $\boldsymbol{Ax} = \boldsymbol{0}$ is a solution of $\boldsymbol{BAx} = \boldsymbol{0}$, i.e.

$$\{\text{solutions of } \boldsymbol{Ax} = \boldsymbol{0}\} \subseteq \{\text{solutions of } \boldsymbol{BAx} = \boldsymbol{0}\}.$$

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c. (iii) Always infinitely many.

d. (i) Exactly one.

Block Matrix Multiplication

Let $\boldsymbol{A}$ be an $(m \times n)$-matrix and let $\boldsymbol{B}$ be an $(n \times p)$-matrix.
For $1 \leq j \leq p$, let $\boldsymbol{b}_j$ denote the j-th column of $\boldsymbol{B}$, so that

$$\boldsymbol{B} = \begin{pmatrix} \boldsymbol{b}_1 & \boldsymbol{b}_2 & \cdots & \boldsymbol{b}_p \end{pmatrix}.$$

Then

$$\boldsymbol{AB} = \boldsymbol{A}\begin{pmatrix} \boldsymbol{b}_1 & \boldsymbol{b}_2 & \cdots & \boldsymbol{b}_p \end{pmatrix} = \begin{pmatrix} \boldsymbol{Ab}_1 & \boldsymbol{Ab}_2 & \cdots & \boldsymbol{Ab}_p \end{pmatrix}.$$

In other words,

$$j\text{-th column of } \boldsymbol{AB} = \boldsymbol{A} \times j\text{-th column of } B.$$

Block Matrix Multiplication

Let $\boldsymbol{A}$ be an $(m \times n)$-matrix and let $\boldsymbol{B}$ be an $(n \times p)$-matrix.
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In other words,

$$j\text{-th column of } \boldsymbol{AB} = \boldsymbol{A} \times j\text{-th column of } B.$$

Question 3

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Let $\boldsymbol{A} = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$ and $\boldsymbol{B} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$.

Find a 4×3 matrix $\boldsymbol{X}$ and a 3×4 matrix $\boldsymbol{Y}$ such that

$$\boldsymbol{AX} = \boldsymbol{I}_3 = \boldsymbol{YB}.$$

Question 3: Solution

Let $\boldsymbol{X} = \begin{pmatrix} \boldsymbol{x}_1 & \boldsymbol{x}_2 & \boldsymbol{x}_3 \end{pmatrix}$. Then $\boldsymbol{AX} = \begin{pmatrix} \boldsymbol{Ax}_1 & \boldsymbol{Ax}_2 & \boldsymbol{Ax}_3 \end{pmatrix}$, so that $\boldsymbol{AX} = \boldsymbol{I}_3$ yields

$$\boldsymbol{Ax}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \ \boldsymbol{Ax}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \ \boldsymbol{Ax}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Solve for $\boldsymbol{x}_1$, $\boldsymbol{x}_2$ and $\boldsymbol{x}_3$ simultaneously:

$$\left(\begin{array}{cccc|c|c|c} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{\text{Gauss-Jordan}} \left(\begin{array}{cccc|c|c|c} 1 & 0 & 0 & 2 & 1 & -1 & 1 \\ 0 & 1 & 0 & -1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$\boldsymbol{x}_1 = \begin{pmatrix} 1-2t_1 \\ t_1 \\ -t_1 \\ t_1 \end{pmatrix}, \ \boldsymbol{x}_2 = \begin{pmatrix} -1-2t_2 \\ 1+t_2 \\ -t_2 \\ t_2 \end{pmatrix}, \ \boldsymbol{x}_3 = \begin{pmatrix} 1-2t_3 \\ -1+t_3 \\ 1-t_3 \\ t_3 \end{pmatrix}, \quad t_1, t_2, t_3 \in \mathbb{R}.$$

Any values of $t_1, t_2, t_3 \in \mathbb{R}$ yield an $\boldsymbol{X}$ satisfying $\boldsymbol{AX} = \boldsymbol{I}_3$.

Question 3: Solution

Let $\boldsymbol{X} = \begin{pmatrix} \boldsymbol{x}_1 & \boldsymbol{x}_2 & \boldsymbol{x}_3 \end{pmatrix}$. Then $\boldsymbol{AX} = \begin{pmatrix} \boldsymbol{Ax}_1 & \boldsymbol{Ax}_2 & \boldsymbol{Ax}_3 \end{pmatrix}$, so that $\boldsymbol{AX} = \boldsymbol{I}_3$ yields

$$\boldsymbol{Ax}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \ \boldsymbol{Ax}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \ \boldsymbol{Ax}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Solve for $\boldsymbol{x}_1$, $\boldsymbol{x}_2$ and $\boldsymbol{x}_3$ simultaneously:

$$\left(\begin{array}{cccc|c|c|c} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{\text{Gauss-Jordan}} \left(\begin{array}{cccc|c|c|c} 1 & 0 & 0 & 2 & 1 & -1 & 1 \\ 0 & 1 & 0 & -1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$\boldsymbol{x}_1 = \begin{pmatrix} 1-2t_1 \\ t_1 \\ -t_1 \\ t_1 \end{pmatrix}, \ \boldsymbol{x}_2 = \begin{pmatrix} -1-2t_2 \\ 1+t_2 \\ -t_2 \\ t_2 \end{pmatrix}, \ \boldsymbol{x}_3 = \begin{pmatrix} 1-2t_3 \\ -1+t_3 \\ 1-t_3 \\ t_3 \end{pmatrix}, \quad t_1, t_2, t_3 \in \mathbb{R}.$$

Any values of $t_1, t_2, t_3 \in \mathbb{R}$ yield an $\boldsymbol{X}$ satisfying $\boldsymbol{AX} = \boldsymbol{I}_3$.

Question 3: Solution (Cont'd)

For $\boldsymbol{Y}$, note that $\boldsymbol{YB} = \boldsymbol{I}_3$ if and only if $\boldsymbol{B}^T\boldsymbol{Y}^T = (\boldsymbol{YB})^T = \boldsymbol{I}_3^T = \boldsymbol{I}_3$. Using the earlier method, we can solve for $\boldsymbol{Y}^T$, and hence for $\boldsymbol{Y}$.

$$\left(\begin{array}{cccc|c|c|c} 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{\text{Gauss-Jordan}} \left(\begin{array}{cccc|c|c|c} 1 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array}\right)$$

$$\boldsymbol{y}_1^T = \begin{pmatrix} \frac{1}{2}-\frac{1}{2}t_1 \\ \frac{1}{2}+\frac{1}{2}t_1 \\ -\frac{1}{2}-\frac{1}{2}t_1 \\ t_1 \end{pmatrix}, \boldsymbol{y}_2^T = \begin{pmatrix} -\frac{1}{2}-\frac{1}{2}t_2 \\ \frac{1}{2}+\frac{1}{2}t_2 \\ \frac{1}{2}-\frac{1}{2}t_2 \\ t_2 \end{pmatrix}, \boldsymbol{y}_3^T = \begin{pmatrix} \frac{1}{2}-\frac{1}{2}t_3 \\ -\frac{1}{2}+\frac{1}{2}t_3 \\ \frac{1}{2}-\frac{1}{2}t_3 \\ t_3 \end{pmatrix}, \quad t_1, t_2, t_3 \in \mathbb{R}.$$

Answer: $\boldsymbol{Y} = \begin{pmatrix} \frac{1}{2}-\frac{1}{2}t_1 & \frac{1}{2}+\frac{1}{2}t_1 & -\frac{1}{2}-\frac{1}{2}t_1 & t_1 \\ -\frac{1}{2}-\frac{1}{2}t_2 & \frac{1}{2}+\frac{1}{2}t_2 & \frac{1}{2}-\frac{1}{2}t_2 & t_2 \\ \frac{1}{2}-\frac{1}{2}t_3 & -\frac{1}{2}+\frac{1}{2}t_3 & \frac{1}{2}-\frac{1}{2}t_3 & t_3 \end{pmatrix}, \quad t_1, t_2, t_3 \in \mathbb{R}.$

Question 3: Solution (Cont'd)

For $\boldsymbol{Y}$, note that $\boldsymbol{YB} = \boldsymbol{I}_3$ if and only if $\boldsymbol{B}^T\boldsymbol{Y}^T = (\boldsymbol{YB})^T = \boldsymbol{I}_3^T = \boldsymbol{I}_3$. Using the earlier method, we can solve for $\boldsymbol{Y}^T$, and hence for $\boldsymbol{Y}$.

$$\left(\begin{array}{cccc|c|c|c} 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{\text{Gauss-Jordan}} \left(\begin{array}{cccc|c|c|c} 1 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array}\right)$$

$$\boldsymbol{y}_1^T = \begin{pmatrix} \frac{1}{2}-\frac{1}{2}t_1 \\ \frac{1}{2}+\frac{1}{2}t_1 \\ -\frac{1}{2}-\frac{1}{2}t_1 \\ t_1 \end{pmatrix}, \boldsymbol{y}_2^T = \begin{pmatrix} -\frac{1}{2}-\frac{1}{2}t_2 \\ \frac{1}{2}+\frac{1}{2}t_2 \\ \frac{1}{2}-\frac{1}{2}t_2 \\ t_2 \end{pmatrix}, \boldsymbol{y}_3^T = \begin{pmatrix} \frac{1}{2}-\frac{1}{2}t_3 \\ -\frac{1}{2}+\frac{1}{2}t_3 \\ \frac{1}{2}-\frac{1}{2}t_3 \\ t_3 \end{pmatrix}, \quad t_1, t_2, t_3 \in \mathbb{R}.$$

Answer: $\boldsymbol{Y} = \begin{pmatrix} \frac{1}{2}-\frac{1}{2}t_1 & \frac{1}{2}+\frac{1}{2}t_1 & -\frac{1}{2}-\frac{1}{2}t_1 & t_1 \\ -\frac{1}{2}-\frac{1}{2}t_2 & \frac{1}{2}+\frac{1}{2}t_2 & \frac{1}{2}-\frac{1}{2}t_2 & t_2 \\ \frac{1}{2}-\frac{1}{2}t_3 & -\frac{1}{2}+\frac{1}{2}t_3 & \frac{1}{2}-\frac{1}{2}t_3 & t_3 \end{pmatrix}, \quad t_1, t_2, t_3 \in \mathbb{R}.$

Question 3: Solution (Cont'd)

For $\boldsymbol{Y}$, note that $\boldsymbol{YB} = \boldsymbol{I}_3$ if and only if $\boldsymbol{B}^T\boldsymbol{Y}^T = (\boldsymbol{YB})^T = \boldsymbol{I}_3^T = \boldsymbol{I}_3$. Using the earlier method, we can solve for $\boldsymbol{Y}^T$, and hence for $\boldsymbol{Y}$.

$$\left(\begin{array}{cccc|c|c|c} 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{\text{Gauss-Jordan}} \left(\begin{array}{cccc|c|c|c} 1 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array}\right)$$

$$\boldsymbol{y}_1^T = \begin{pmatrix} \frac{1}{2}-\frac{1}{2}t_1 \\ \frac{1}{2}+\frac{1}{2}t_1 \\ -\frac{1}{2}-\frac{1}{2}t_1 \\ t_1 \end{pmatrix}, \boldsymbol{y}_2^T = \begin{pmatrix} -\frac{1}{2}-\frac{1}{2}t_2 \\ \frac{1}{2}+\frac{1}{2}t_2 \\ \frac{1}{2}-\frac{1}{2}t_2 \\ t_2 \end{pmatrix}, \boldsymbol{y}_3^T = \begin{pmatrix} \frac{1}{2}-\frac{1}{2}t_3 \\ -\frac{1}{2}+\frac{1}{2}t_3 \\ \frac{1}{2}-\frac{1}{2}t_3 \\ t_3 \end{pmatrix}, \quad t_1, t_2, t_3 \in \mathbb{R}.$$

Answer: $\boldsymbol{Y} = \begin{pmatrix} \frac{1}{2}-\frac{1}{2}t_1 & \frac{1}{2}+\frac{1}{2}t_1 & -\frac{1}{2}-\frac{1}{2}t_1 & t_1 \\ -\frac{1}{2}-\frac{1}{2}t_2 & \frac{1}{2}+\frac{1}{2}t_2 & \frac{1}{2}-\frac{1}{2}t_2 & t_2 \\ \frac{1}{2}-\frac{1}{2}t_3 & -\frac{1}{2}+\frac{1}{2}t_3 & \frac{1}{2}-\frac{1}{2}t_3 & t_3 \end{pmatrix}, \quad t_1, t_2, t_3 \in \mathbb{R}.$

Next Lesson ...

We will explore **Sections 2.4–2.6**.
Please watch the video clips and review the lecture slides for these sections before then.